

Dr. Sumit Kumar Gupta

Assistant Professor, Department of Computer Science & Engineering
Indian Institute of Information Technology (IIIT) Pune
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CAREER OBJECTIVE:

To contribute to teaching, research, and academic administration in a premier institution by effectively utilizing my technical expertise, research experience, and leadership skills for the holistic development of students and institutional growth.

PROFESSIONAL EXPERIENCE:

- Assistant Professor Department of Computer Science & Engineering
Additional Administrative Responsibility: Assistant Registrar (Academics),
Indian Institute of Information Technology (IIIT) Pune
July 2024 – Present
- Assistant Professor, Department of Computer Science & Engineering
Maulana Azad National Institute of Technology (MANIT), Bhopal, M.P.
August 2022 – June 2024
- Research Scholar, Department of Computer Science & Engineering
Maulana Azad National Institute of Technology (MANIT), Bhopal, M.P.
August 2017 – August 2022
- Assistant Professor, Department of Computer Science & Engineering
Lakshmi Narain College of Technology Excellence, Bhopal, M.P.
July 2015 – August 2017
- Assistant Professor, Department of Computer Science & Engineering
University Institute of Technology, RGPV, Bhopal, M.P.
February 2015 – June 2015
- Research Scholar, Department of Computer Science & Engineering
Maulana Azad National Institute of Technology (MANIT), Bhopal, M.P.
August 2012 – June 2014
- Assistant Professor, Department of Computer Science & Engineering
Lakshmi Narain College of Technology, Bhopal, M.P.
June 2011 – July 2012

SIGNIFICANT ACHIEVEMENTS:

I have secured valid UGC NET scorecards in June 2015 & July 2018, whose descriptions mentioned below are with roll number

EXAM	YEAR	ROLL NO.	CUT OFF	PERCENTAGE
UGC NET	2015	39001181	62.38	65.14
UGC NET	2018	38001613	51.33	51.33

I have secured valid GATE scorecards from 2011 to now(2024), whose description is mentioned below with the Registration number.

EXAM	YEAR	REGISTRATION NUMBER	GATE SCORE	QUALIFYING MARKS	MARKS OBTAINED
GATE	2024	CS24S55012502	613	27.6	51.76
GATE	2023	CS23S15012666	465	32.5	42.67
GATE	2022	CS22S15014330	420	25	30.33
GATE	2021	CS21S55013227	439	26.1	33.39
GATE	2020	20S65010161	488	28.5	40.33
GATE	2019	19S35012694	352	29.5	29.67
GATE	2018	18S35014138	445	25.0	33.00
GATE	2017	17S505011030	496	25.0	38.02
GATE	2016	16S65022065	463	25.0	35.09
GATE	2015	55021S7042	493	25.0	36.78
GATE	2014	500307327	484	25.0	34.97
GATE	2013	51921255	424	25.0	29.33
GATE	2012	5203035	589	31.5	52.00
GATE	2011	5227283	355	25.0	27.33

ACADEMIC QUALIFICATION:

Degree	College/School	Board / University	Passing Year	Percentage or C.G.P.A.
Ph.D. (CSE)	Maulana Azad National Institute Of Technology, Bhopal.	MANIT, Bhopal	2023	8.79 (Course Work)
M.Tech (CSE)	Maulana Azad National Institute Of Technology, Bhopal.	MANIT, Bhopal	2014	9.18

B.E. (CSE)	Jai Narain College Of Technology, Bhopal.	RGPV, Bhopal	2011	73.66
Higher Secondary	Demonstration Multipurpose School, Bhopal.	CBSE	2007	70.00
High School	Demonstration Multipurpose School, Bhopal.	CBSE	2005	60.40

RESEARCH CREDENTIALS:

- Google Scholar details:
<https://scholar.google.com/citations?user=79vGSWoAAAAJ&hl=en>
- ORCID ID :
<https://orcid.org/0000-0002-8550-2194>
- SCOPUS ID :
Author ID: 57217374270

PUBLICATION:

SCIE Journals Publications

- Likhar, P., Gupta, S. K., Choudhary, J., & Singh, D. P. (2026). Delving deep: DDoS attack resilience through deep learning approaches. Knowledge and Information Systems, 68(1), 19. (Published in SCIE(Q1)).
- Gupta, S. K., Singh, D. P., and Choudhary, J. (2022). A review of clique-based overlapping community detection algorithms. Knowledge and Information Systems: vol 64, pp 2023-2058. (Published in SCIE(Q1)).
- Gupta, S.K., Singh, D.P. (2022). Seed Community Identification Framework for Community Detection over Social Media. Arabian Journal for Science and Engineering: vol 48, pp 1829–1843. (Published in SCIE(Q1)).
- Gupta, S., and Singh, D. P. (2020). Recent trends on community detection algorithms: A survey. Modern Physics Letters B, vol 34, 2050408(35), DOI: 10.1142/S0217984920504084. (Published in SCIE).

Scopus Indexed Publications

- Kumar S., Rahangdale S.T., and Gupta S.K. (2026). Efficient Probabilistic Forecasting of Solar and Wind Generation Using a GAT–Adapt–Hybrid Diffusion Model. Accepted in IEEE International Conference on Interdisciplinary Approaches in Technology and Management for Social Innovation (IATMSI-2026)

- Gupta, S., and Singh, D. P. (2020). Greedy Graph Coloring Algorithm based on Depth First Search. International Journal on Emerging Technologies, 11(2): 854–862. (Published in SCOPUS INDEXED JOURNAL).
- Mansuri, S., Gupta, S. K., Singh, D. P., and Choudhary, J.(2022). Modified DMGC algorithm using ReLU-6 with improved learning rate for complex cluster associations. International Conference on Intelligent Computing and Control Systems, IEEE, PP. 634-640(Published in SCOPUS INDEXED CONFERENCE).
- Gupta, S. K., and Singh, D. P. (2023). CBLA: A Clique Based Louvain Algorithm for Detecting Overlapping Community. Procedia Computer Science, vol 218, pp 2201-2209 (Published in SCOPUS INDEXED JOURNAL).
- Gupta, S. K., Singh, D. P., and Choudhary, J. (2023). New GPU Sorting Algorithm Using Sorted Matrix. Procedia Computer Science, vol 218, pp 1682-1691(Published in SCOPUS INDEXED JOURNAL).

Book / Book Chapter Publications

- Gupta, S., Nandanwar, A.K., and Singh, D. P. (2021). DATA STRUCTURE. Publisher: LAP LAMBERT Academic Publishing, Paperback: 188 pages, ISBN-10: 6204715909, ISBN-13: 978-6204715902.
- Gupta, S. K., Yadav, S., Singh, D. P., and Choudhary, J. (2022). Local Roughness Binary Pattern for Texture Classification. In Lecture Notes in Electrical Engineering book series (LNEE,volume 907). ISBN 978-981-19-4686-8

International Journal Publications

- Gupta, S., and Rasool, A. (2014). Bit Parallel String-Matching Algorithms: A Survey. International Journal of Computer Application, 95(10).
- Gupta, S., Gupta, N., and Gupta, R. (2014). Objects and Method Calling in Java Virtual Machine. International Journal of Computer Application, 86(11).
- Bee, S., and Gupta, S. (2016). A Brief Survey of Various Approaches For Feature of Text Mining. International Journal of Research in Computer Applications and Robotics, 4(7).
- Bee, S., and Gupta, S. (2016). A Genetic Approach to Classify Text Documents by Utilizing Term Features. International Journal of Advanced Research in Computer Engineering & Technology, 5(12): 2762-2768.
- Kaushik, S., and Gupta, S. (2017). Deployment of OpenStack using DevOps. International Journal of Computer Science & Communication Networks, 7(3).

PATENT:

Australian Patent (Granted)

Patent No.: 2021104850

Title: IoT-based system and method for monitoring traffic and controlling traffic signals using machine learning

Publication & Grant Date: 02 June 2022

MENTORING & STUDENT SUPERVISION:

Doctoral Students Supervised as Primary Advisor

- Juhi Agrawal(312415001), Ph.D. in Computer Science & Engineering (2024 – present), Indian Institute of Information Technology, Pune, Thesis Topic: A Hybrid Recommendation System Using SVD, Genre Embeddings, and ANN to Alleviate Data Sparsity.
- Seema Arya(322515002), Ph.D. in Computer Science & Engineering (2025 – present), Indian Institute of Information Technology, Pune, Thesis Topic: Deep Learning.
- Sushil Kumar Mishra(312515008), Ph.D. in Computer Science & Engineering (2025 – present), Indian Institute of Information Technology, Pune, Thesis Topic: Security using Machine Learning.

Masters Students Supervised (Thesis)

- Sakshi Kaushik, M.Tech. in Computer Science & Engg. (completed: 2017)
Thesis: Implementation of Open Stack Through Ansible.
- Sharda Dubey, M.Tech. in Computer Science & Engg. (completed: 2017)
Thesis: An Innovative Method for Detecting and Preventing Against ARP and IP Spoofing.

Masters Students Supervised (Project)

- Ricky Kumar (212515019), M.Tech. in Computer Science & Engg. (Dec 2025)
Project: Hybrid Phishing Detection Using Static and Dynamic Behavioral Analysis.
- Samarpita Garai (212515012), M.Tech. in Computer Science & Engg. (Dec 2025)
Project: Insurance Fraud Detection using Machine Learning.

- Piyush Khanke (222415014), M.Tech. in Computer Science & Engg. (June 2025)
Project: MediGenius: AI Powered Prescription Guide.
- Vidhi Srivastava (222415022), M.Tech. in Computer Science & Engg. (June 2025)
Project: Automated Software Testing Using Generative AI.
- Ganesh Kini (222415007), M.Tech. in Computer Science & Engg. (Dec 2024)
Project: Enhance Fairness in Online Exam Proctoring: Reducing false positive with YOLOv5 and Deep Learning.
- Lomesh Patil (222415010), M.Tech. in Computer Science & Engg. (Dec 2024)
Project: Human Activity Recognition based on LRCN and ConvLSTM.

Bachelor's Students Supervised (Project)

- Soyal Rahangdale (112215175), B.Tech. in Computer Science & Engg. (Dec 2025)
Project: Efficient Probabilistic Forecasting of Solar and Wind Generation Using a GAT-Adapt-Hybrid Diffusion Model.
- Sripathi Praneeth (112215176), B.Tech. in Computer Science & Engg. (Dec 2025)
Project: GomuNet: The XML-Powered Mind-Reader for Gomoku Moves using Deep Learning.
- Sudhanshu Kumar(112215178), B.Tech. in Computer Science & Engg. (Dec 2025)
Project: GAT-Adaptive-Hybrid Diffusion Framework for Probabilistic Solar and Wind Forecasting.
- Shweta Gaikwad (112315182), Gandhale Vilas (112315062), Sandhya Samrath (112316040) & Anamika meena (112316005), B.Tech. in Computer Science & Engg / Electronics and Communication Engineering . (Dec 2025) Project: Stacked Ensemble Learning with SMOTE-ENN and K-Means Clustering for Diabetes Prediction.
- Aman Kumar Shah (112315018), Ayush Tiwari (112315039), Aman (112315017), Burhanuddin Mohammadi Gulamaliwala (112315044) & Madhur Sunhare (112316025), B.Tech. in Computer Science & Engg. (Dec 2025) Project: Secure and Scalable Federated Learning for Multi-Institutional Healthcare Data.
- Archit Gulve (112315026), Akshad Khune (112315086), Atharva Khedekar (112315035), Avani Patil (112315127) & Swaraj Raskar (112315150), B.Tech. in

Computer Science & Engg. (Dec 2025) Project: Modeling Epidemic Spread In Evolving Social Graphs Using Temporal GNN.

- Luv Kansal (112315094), Manaswi Rajne (112315101), Bhoomi Gundecha (112315067), Aryan Gupta (112315033) & Kunj Vipul Goyal (112315091), B.Tech. in Computer Science & Engg. (Dec 2025) Project: SatyaShield- AI Powered Misinformation Combatter.
- Subhiksha Manivannan (112315096), Nikhil VGK (112316031), Nikita Prasad (112315118), Netra (112316030) & Yubhashna Shinde (112315214), B.Tech. in Computer Science & Engg / Electronics and Communication Engineering . (Dec 2025) Project: Time Fusion Transformers: Integrating Deep sentiment Embedding for Probabilistic E-Commerce Sales Forecasting.
- Kedar Ekmode (112115072), Devesh Raghuvanshi (112115042), Aman Garg (112115019) & Pratham Dhadhiya (112115114), B.Tech. in Computer Science & Engg. (June 2025) Project: DDOS Attack: Detection and Mitigation.
- Nimesh Mali (112115083), B.Tech. in Computer Science & Engg. (June 2025) Project: Interactive Sorting Algorithm Visualizer and Comparator.
- Aaroh Maheshwari (112116001), B.Tech. in Computer Science & Engg. (June 2025) Project: Health Non-Fungible Tokens(NFT)- Medmask.
- Arjun Anil Sagar (112215152), Atharva Rajendra Padghan (112216009), Prajwal Yuvraj Khadse (112215140) & Prathmesh Pradip Dode (112215141), B.Tech. in Computer Science & Engg / Electronics and Communication Engineering. (June 2025) Project: Genetic Disease Detection using Deep Learning.
- Sarvagya Gupta (112115140), B.Tech. in Computer Science & Engg. (June 2025) Project: Loyalty Program Using Blockchain.
- Pratham Dhadhiya (112115114), B.Tech. in Computer Science & Engg. (June 2025) Project: Healthify: AI Powered Medical Laboratory System.
- Devesh Raghuvanshi (112115042), B.Tech. in Computer Science & Engg. (June 2025) Project: Anonymization based Privacy Preservation.
- Dhruv Singla (112115045), B.Tech. in Computer Science & Engg. (June 2025) Project: Book Hive.

- Arjun Anil Sagar (112215152), Atharva Rajendra Padghan (112216009), Prajwal Yuvraj Khadse (112215140), Anushka Raut (112216038) & Prathmesh Pradip Dode (112215141), B.Tech. in Computer Science & Engg. (June 2025) Project: Breast Cancer Detection using Deep Learning.
- Som Singh Lodhi (112115157), B.Tech. in Computer Science & Engg. (June 2025) Project: A Comparative Analysis of Encryption Methods.

PROJECT:

- PhD thesis work title is “**EFFICIENT COMMUNITY DETECTION ALGORITHMS FOR LARGE NETWORKS**” under which we have implemented three different algorithms for finding overlapping communities from the given graph.
- Dissertation in M.Tech on “**PERFORMANCE IMPROVEMENT OF VARIOUS BIT PARALLEL ALGORITHMS,**” which are the algorithms for string matching where we have to find the pattern from the text.
- Major Project on “**INTRUDER DETECTION SYSTEM,**” is used to prevent intruders from entering the system and taking advantage of it.
- Minor Project on “**STEGNOGRAPHY,**” Which is used to hide the message into the image and audio with the help of the DES algorithm.

TECHNICAL SKILLS:

- Programming Language: C, C++, PyTorch, TensorFlow & Python.
- Favorite Subjects: TOC, Data Structure & Algorithms, Compiler.
- Research Areas: Graph Mining, Community Detection, ML Security, Deep Learning
- Comfortable working on the OS environment.

WORKSHOP / SEMINAR ATTENDED:

- One week Faculty Development Programme on Big Data & Hadoop Organized by Department of Information Technology UIT, RGPV Sponsored by TEQIP II and Technically associated with CSI and ACM.

- One week Faculty Development Programme on Use of Modern Computing Techniques for Testing and Web Designing Organized by RGPV under TEQIP III in association with Lakshmi Narain College of Technology, Bhopal.
- Two-days National Seminar on Digital India Towards Innovation Held at Lakshmi Narain College of Technology Excellence, Bhopal Sponsored by M.P. Council of Science and Technology Bhopal, CSI Bhopal Chapter and ACM Bhopal Chapter.
- One week Workshop on Application of Deep Learning and Generative AI in Computer Vision and NLP Task Organized by Department of Computer Science and Engineering, Maulana Azad National Institute of Technology Bhopal.
- One week Faculty Development Programme on Natural Language Processing jointly Organized by Electronics and ICT Academics held at Maulana Azad National Institute of Technology Bhopal.
- One week short term training programme on Soft Computing & Machine Learning conducted by Department of Computer Science and Engineering, Maulana Azad National Institute of Technology Bhopal.
- One week short term training programme on Hands-on Experience in Big Data Analysis Using Hadoop Organized by Department of Computer Science and Engineering, Maulana Azad National Institute of Technology Bhopal.
- Two-days workshop on the Development of Web-based Data Repository Organized by the Department of Computer Science and Engineering, Maulana Azad National Institute of Technology Bhopal Sponsored by TEQIP II.

REVIEWER AND SESSION CHAIR:

- Chaired a Session in IEEE 1st International Conference on Data Science and Intelligent Network Computing(ICDSINC 2025) held at National Institute of Technology Raipur.
- Chaired a Session in 7th International Conference on Recent Trends in Image Processing & Pattern Recognition (RTIP2R) held at Indian Institute of Information Technology, Bhopal in collaboration with Maulana Azad National Institute of Technology Bhopal, India and 2AI: Applied AI Research Lab - University of South Dakota (USA).
- Jury Member in a Session in 10th IEEE International Students' Conference on Electrical, Electronics and Computer Science (SCEECS 2025) held at Maulana Azad National Institute of Technology Bhopal.
- Participated in Capacity Building on Specific Learning Disabilities (SLD) Cycle 2 with office of student life or campus life affairs organized by National Institute of Educational Planning and Administration, Delhi.

- Served as Reviewer at IEEE International Conference on Advanced Computation & Telecommunication 2018 held at Lakshmi Narain College of Technology, Bhopal.
- Served as Reviewer at IEEE International Conference on Artificial Intelligence and Emerging Technologies (ICAIET-2025) held at School of Computer Science and Engineering, XIM University, Bhubaneswar.
- Served as Reviewer at International Conference on Big data, Cloud Computing & Intelligent System 2019 at Lakshmi Narain College of Technology, Bhopal.
- Served as Reviewer at IEEE Delhi Section International Conference on Integrated Circuits, Communication, and Computing Systems (ICIC3S-2024), organized by School of Electronics, Indian Institute of Information Technology Una.
- Served as Reviewer at IEEE MP Section International Conference on Engineering Innovations and Technologies (ICoEIT 2025) held at Lakshmi Narain College of Technology, Bhopal.
- Served as Reviewer at IEEE International Conference on Electrical, Electronics and Computer Science with Advance Power Technologies – A Future Trend (ICE2CPT 2025), organized by the Department of Electrical Engineering, National Institute of Technology Jamshedpur, in association with the IEEE Kolkata Section, IEEE Industrial Electronics Society and IEEE Student Branch, NIT Jamshedpur.

EXPERT LECTURE DELIVERED / PAPER PRESENTER:

- Delivered a expert lecture on “Significance of Community Detection Algorithms in Real World Application” at G H Rasoni College of Engineering and Management, Pune.
- Delivered a expert lecture on “Approach to Design a PPT for Smart India Hackathon” organized by Department of Computer Science and Engineering, Sagar Institute of Science & Technology, Bhopal.
- Delivered a expert lecture on “Mastering Python: Beyond the Basics into Real World Applications” organized by Department of Computer Science and Engineering, Sagar Institute of Science & Technology, Bhopal.
- Presented paper titled “New GPU sorting algorithm using sorted matrix” in ICMLDE conference held at UPES, Dehradun.
- Presented paper titled “CBLA: A Clique Based Louvain Algorithm for Detecting Overlapping Community” in ICMLDE conference held at UPES, Dehradun.
- Presented paper titled “New GPU sorting algorithm using sorted matrix” in ICMIT conference held at Amity University, Jharkhand.

- Presented paper titled “Evolution of new version of internet protocol (IPv6): Replacement of IPv4” in National Conference on Emerging Trends in Information Technology IITM Delhi.
- Presented paper titled “Local roughness binary pattern for texture classification” in 3rd springer international Conference on Data Engineering & Applications held at State Technical University, Bhopal.

PROFESSIONAL MEMBERSHIP:

- Member of Institute of Electrical and Electronics Engineers (IEEE).
- Member of the Editorial Board for the Journals of Lattice Science Publication (LSP).
- Member of the Editorial Board for the Journals of Blue Eyes Intelligence Engineering and Sciences Publication (BEIESP).
- Member of the Editorial Board for the Journals of International Journal of Innovative Research in Technology and Management.

OTHER ACTIVITIES & ACHIEVEMENTS:

- Topper of M.Tech CSE in the second year and secured second position in overall.
- Secured First Position in Tug of War at the Inter-IIIT Faculty Sports Meet held at ABV-IIITM Gwalior.
- Won Individual Gold Medal in Shot Put at the Inter-IIIT Faculty Sports Meet held at ABV-IIITM, Gwalior.
- Secured Third Position in Cricket at the Inter-IIIT Faculty Sports Meet held at ABV-IIITM Gwalior.
- Mentor of the winning team at the Smart India Hackathon held at IIT Kharagpur.
- Served as External Examiner for the Ph.D. thesis entitled ‘Random Key Based Image Cryptographic System with Computational Efficiency in a Dynamic Network’ at Rabindranath Tagore University, Bhopal.
- Served as External Expert (Examiner) for the M.Tech Final Thesis Presentations at the Department of CSE, Maulana Azad National Institute of Technology.
- Served as External Examiner for the Project-II End-Term Examination of B.Tech (Cloud Technology & Information Security) and MCA (Information Security) on 22nd May 2025 at Ajeenkya D.Y. Patil University, Pune.

- Coordinator of Capacity Building for promoting positive mental health, Resilience and wellbeing in HEI's under the Malaviya Mission Teacher Training Program(MMTTP).
- Served as a Chairman SRC Committee of Ph.D. Scholar Megha Purohit student of Computer Science and Engineering Department of Indian Institute of Information Technology Pune.
- Board of Studies (BOS) Member of Computer Science and Engineering Department of Indian Institute of Information Technology Pune.

PERSONAL PROFILE:

- Father Name : Magan Bihari Gupta
- Occupation : Retd. Deputy ranger in Forest Department Bhopal
- Mother Name : Shashi Gupta
- Category : General
- Marital Status : Married
- Date Of Birth : 01-May-1989

DECLARATION:

I hereby declare that the information mentioned above is correct up to my knowledge, and I bear the responsibility for the correctness of the particulars mentioned above.

Date:

Place:

(Dr. Sumit Kumar Gupta)